

Heaven's light is our guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering



Course Code: CSE 3102

Course Name: Database Systems Sessional

Lab Work title: Create Database & Quarries like given Lab Exercise, and Solve that

Lab No: 10

Date of Lab: 16 June 2025

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Flight Reservation System

1. Create a database called "FlightReservationDB". Write the corresponding query for this in your report.

Solution: *Create the database.*

```
CREATE DATABASE FlightReservationDB;
```

2. Create a table titled "Passengers" with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
passenger_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
full_name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	UNIQUE, NOT NULL
phone_number	VARCHAR(15)	NOT NULL
date_of_birth	DATE	NOT NULL
nationality	VARCHAR(50)	NOT NULL

Solution: *Create the "Passengers" table.*

```
CREATE TABLE Passengers (  
    passenger_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
    full_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    phone_number VARCHAR(15) NOT NULL,  
    date_of_birth DATE NOT NULL,  
    nationality VARCHAR(50) NOT NULL  
);
```

3. Create a table titled "Flights" with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
flight_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
flight_number	VARCHAR(10)	UNIQUE, NOT NULL
departure_city	VARCHAR(50)	NOT NULL
arrival_city	VARCHAR(50)	NOT NULL
departure_time	DATETIME	NOT NULL
arrival_time	DATETIME	NOT NULL

Solution: *Create the "Flights" table*

```
CREATE TABLE Flights (  
    flight_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
    flight_number VARCHAR(10) UNIQUE NOT NULL,  
    departure_city VARCHAR(50) NOT NULL,  
    arrival_city VARCHAR(50) NOT NULL,  
    departure_time DATETIME NOT NULL,  
    arrival_time DATETIME NOT NULL  
);
```

4. Create a table titled "Bookings" with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
booking_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
passenger_id	INT	NOT NULL
flight_id	INT	NOT NULL
booking_date	DATE	NOT NULL
seat_number	VARCHAR(5)	NOT NULL
class	VARCHAR(20)	NOT NULL

Solution: Create the "Bookings" table

```
CREATE TABLE Bookings (  
    booking_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
    passenger_id INT NOT NULL,  
    flight_id INT NOT NULL,  
    booking_date DATE NOT NULL,  
    seat_number VARCHAR(5) NOT NULL,  
    class VARCHAR(20) NOT NULL,  
    FOREIGN KEY (passenger_id) REFERENCES  
Passengers (passenger_id),  
    FOREIGN KEY (flight_id) REFERENCES Flights (flight_id)  
);
```

5. Populate the "Passengers" table with the following data. Write the corresponding insertion query for this in your report.

full_name	email	phone_number	date_of_birth	nationality
Alice Johnson	alice@gmail.com	+11234567890	1990-03-12	USA
Ravi Sharma	ravi@outlook.com	+919876543210	1985-07-25	India
Sara Lee	sara.lee@yahoo.com	+440712345678	1992-11-08	UK
Muhammad Fahad	fahad@mail.com	+923001112223	1988-01-16	Pakistan
Liu Wei	liu.wei@china.cn	+8613900000001	1995-05-20	China

Solution: *Insert into "Passengers"*

```
INSERT INTO Passengers (full_name, email, phone_number,
date_of_birth, nationality) VALUES
('Alice Johnson', 'alice@gmail.com', '+11234567890', '1990-03-12',
'USA'),
('Ravi Sharma', 'ravi@outlook.com', '+919876543210', '1985-07-25',
'India'),
('Sara Lee', 'sara.lee@yahoo.com', '+440712345678', '1992-11-08',
'UK'),
('Muhammad Fahad', 'fahad@mail.com', '+923001112223',
'1988-01-16', 'Pakistan'),
('Liu Wei', 'liu.wei@china.cn', '+8613900000001', '1995-05-20',
'China');
```

6. Populate the "Flights" table with the following data. Write the corresponding insertion query for this in your report.

flight_number	departure_city	arrival_city	departure_time	arrival_time
FL123	New York	London	2024-07-10 08:00:00	2024-07-10 20:00:00
FL456	Delhi	Dubai	2024-07-11 14:30:00	2024-07-11 17:30:00
FL789	Karachi	Jeddah	2024-07-12 06:00:00	2024-07-12 09:15:00
FL101	Beijing	Sydney	2024-07-15 22:00:00	2024-07-16 10:00:00
FL124	London	New York	2024-07-20 09:00:00	2024-07-20 19:00:00

Solution: *Insert into "Flights"*

```
INSERT INTO Flights (flight_number, departure_city, arrival_city,  
departure_time, arrival_time) VALUES  
( 'FL123', 'New York', 'London', '2024-07-10 08:00:00', '2024-07-10  
20:00:00'),  
( 'FL456', 'Delhi', 'Dubai', '2024-07-11 14:30:00', '2024-07-11  
17:30:00'),  
( 'FL789', 'Karachi', 'Jeddah', '2024-07-12 06:00:00', '2024-07-12  
09:15:00'),  
( 'FL101', 'Beijing', 'Sydney', '2024-07-15 22:00:00', '2024-07-16  
10:00:00'),  
( 'FL124', 'London', 'New York', '2024-07-20 09:00:00', '2024-07-20  
19:00:00');
```

7. Populate the "Bookings" table with the following data. Write the corresponding insertion query for this in your report.

passenger_id	flight_id	booking_date	seat_number	class
1	1	2024-06-01	12A	Economy
2	2	2024-06-02	5C	Business
3	1	2024-06-03	14B	UK
4	3	2024-06-04	2D	Economy
5	4	2024-06-05	21E	Economy
1	5	2024-06-01	15A	Economy

Solution: *Insert into "Bookings"*

```
INSERT INTO Bookings (passenger_id, flight_id, booking_date,
seat_number, class) VALUES
(1, 1, '2024-06-01', '12A', 'Economy'),
(2, 2, '2024-06-02', '5C', 'Business'),
(3, 1, '2024-06-03', '14B', 'Economy'),
(4, 3, '2024-06-04', '2D', 'Business'),
(5, 4, '2024-06-05', '21E', 'Economy'),
(1, 5, '2024-06-01', '15A', 'Economy');
```

After creating and populating the tables, do these **queries**.

- 1) **Query 1** (Easy): List all passengers who are from the USA.

Solution:

```
SELECT *  
FROM Passengers  
WHERE nationality = 'USA';
```

Output:

passenger_id	full_name	email	phone_number	date_of_birth	nationality
1	Alice Johnson	alice@gmail.com	+11234567890	1990-03-12	USA

- 2) **Query 2** (Easy): Retrieve all bookings made before June 4, 2024.

Solution:

```
SELECT *  
FROM Bookings  
WHERE booking_date < '2024-06-04';
```

Output:

booking_id	passenger_id	flight_id	booking_date	seat_number	class
1	1	1	2024-06-01	12A	Economy
2	2	2	2024-06-02	5C	Business
3	3	1	2024-06-03	14B	Economy
6	1	5	2024-06-01	15A	Economy

- 3) **Query 3** (Medium): List the flight numbers and departure cities of flights that arrive in 'London'.

Solution:

```
SELECT flight_number, departure_city  
FROM Flights  
WHERE arrival_city = 'London';
```

Output:

flight_number	departure_city
FL123	New York

- 4) **Query 4** (Medium): Find the full name and email of passengers who booked Business class.

Solution:

```
SELECT P.full_name, P.email
FROM Passengers P
JOIN Bookings B ON P.passenger_id = B.passenger_id
WHERE B.class = 'Business';
```

Output:

full_name	email
Ravi Sharma	ravi@outlook.com
Muhammad Fahad	fahad@mail.com

- 5) **Query 5** (Medium-Hard): Count how many passengers are booked on each flight.

Solution:

```
SELECT F.flight_number, COUNT(B.booking_id) AS
total_booked_passengers
FROM Flights F
JOIN Bookings B ON F.flight_id = B.flight_id
GROUP BY F.flight_number;
```

Output:

flight_number	total_booked_passengers
FL101	1
FL123	2
FL124	1
FL456	1
FL789	1

- 6) **Query 6** (Medium-Hard): Find the most recently booked passenger (i.e., latest booking date).

Solution:

```
SELECT P.full_name, B.booking_date
FROM Passengers P
JOIN Bookings B ON P.passenger_id = B.passenger_id
ORDER BY B.booking_date DESC
LIMIT 1;
```

Output:

full_name	booking_date
Liu Wei	2024-06-05

- 7) **Query 7** (Medium-Hard): List passengers who have booked more than one flight.

Solution:

```
SELECT P.full_name, COUNT(B.booking_id) AS number_of_bookings
FROM Passengers P
JOIN Bookings B ON P.passenger_id = B.passenger_id
GROUP BY P.passenger_id
HAVING COUNT(B.booking_id) > 1;
```

Output:

full_name	number_of_bookings
Alice Johnson	2